

Qatar-1b

R-2713

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_120915 · created 2026-09-01T01:20:36+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456185.82251 +/- 0.00013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.138 +/- 0.012
Transit depth (Rp/Rs)^2	1.92 +/- 0.35 [%]
Orbital Inclination (inc)	85.57 +/- 1.01
Airmass coefficient 1 (a1)	1.345 +/- 0.021
Airmass coefficient 2 (a2)	-0.1774 +/- 0.0082
Scatter in the residuals of the lightcurve fit is	1.62 %
Best Comparison Star	None
Optimal Aperture	3.31
Optimal Annulus	11.45
Transit Duration (day)	0.0747 +/- 0.004

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.138 $\pm$ 0.012 vs 0.1463 (deviation 0.6 σ)
Duration fit vs published	0.0747 d vs 0.0692 d (deviation 1.2 σ)
Mid-time O–C	0.3 min
Depth SNR	10.0
Residual scatter	1.62 %

Light curve

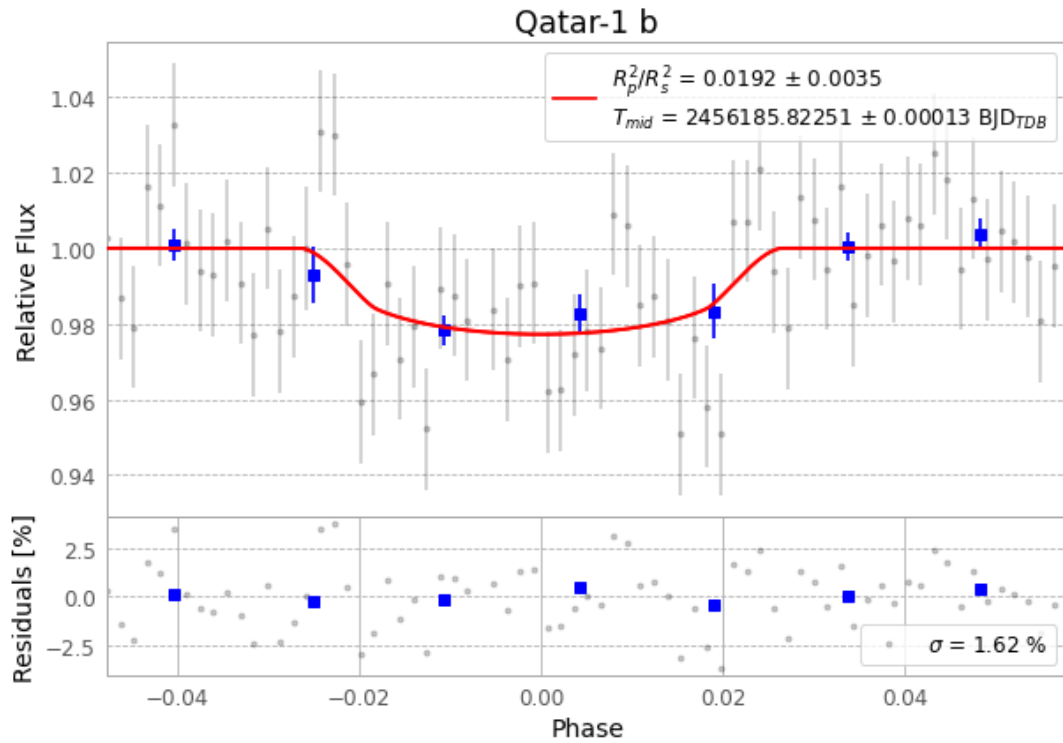


Figure 1 — FinalLightCurve_Qatar-1 b_2012-09-14.png (SHA-256 24ca4f5eea065d27...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [297, 300], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 178a41a00dbf3e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

73 FITS frames (48 MB), Qatar-1120915060314.FITS → Qatar-1120915093912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-120915.log (435dc939990e...), FinalLightCurve_Qatar-1 b_2012-09-14.png (24ca4f5eea06...), FinalLightCurve_Qatar-1 b_2012-09-14.csv (fb325d1adbdb...)

Compute resources

Wall time 133.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-01 01:20Z.